GEOMETRY

ITEM ONE

A certain town council wants to reduce speeding and improve road safety by building a roundabout in the trading Centre. The council wishes not to demolish three existing land marks, the market at **M** (-19,2), the Petro station at **P** (3,24) and the Boda stage at **B** (25,2). The chairperson of the town was tasked with visiting the division engineer's office to seek technical approval for the project.

To combine the trip with the personal health goal, the chairperson decided to cycle to the division engineer's office. His bicycle has a crank arm and the crank Centre is 28cm above the ground. The vertical height of the right pedal from the ground is modeled by $\boldsymbol{h = 28 + 15cos\theta + 8sin\theta}$. He wishes to continue with this exercise if he meets his weight – loss target which is possible if the pedal height varies between **11cm** and **45cm** throughout each rotation.

After listening to the council's proposal, the division engineer set the following conditions before approval is granted:

1. The Centre of the roundabout must lie on the line $\boldsymbol{4y = 3x - 1}$ which represents the existing road and the radius must be at least 15m to allow long vehicles turn safely.
2. A minimum clearance of **5m** must be left from each of the existing land marks to avoid demolition.
3. A road from another town represented by the line $\boldsymbol{y - 10x - 130 = 0}$ must connect to the round about and the distance from the point where this road meets the roundabout should be approximately 10m from the market.

Task:

Using mathematical justification, guide the chairperson and the town council on the decisions to take.

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ITEM TWO

A certain coffee factory is surveying coffee plantation in a certain hilly area. The field team has collected data for mapping the boundary, setting up drip lines and checking the terrain slope.

The boundary of the plantation has been surveyed as an ellipse with equation $4x^2 - 24x + 9y^2 + 18y + 9 = 0$. Where x and y are distances in tens of metres.The field team wishes to put water tank at the centre of the plantation if the distance from the centre to any point on the boundary where the ellipse cuts the semi - major axis is less than 25 m otherwise there will be a need for booster pumps.

The team also wishes to prevent soil erosion if the eccentricity is less than 0.8 otherwise it may incur extra costs of building terraces.

Two straight drip lines $L_1: \begin{pmatrix}2\\0\\1\end{pmatrix} + \beta\begin{pmatrix}1\\2\\-2\end{pmatrix}$ and $L_2: \begin{pmatrix}0\\3\\0\end{pmatrix} + \mu\begin{pmatrix}2\\-1\\2\end{pmatrix}$ are to be installed underground to deliver water to the tank. The team wishes to adjust these lines for maintenance if the angle between them is not in the range $60°$ and $90°$.

The field team also wishes to measure the angle of slope of a hillside section. Three horizontal forces $cos\theta, cos3\theta$ and $cos5\theta$ act parallel to the slope. For stability the sum of these forces is zero. Landslide risk is high for $\theta \geq 100°$.

The team wishes to suspend the project if the number of unsafe angles exceeds the safe angles in the range $0° \leq \theta \leq 180°$

Task:

Based on mathematical justifications, advise the coffee factory on the decisions to take.

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ITEM THREE

A farmer is installing a new sprinkler at point $S(2,1,5)$ inside a green house. The green house roof is the plane $x + 2y - z = 4$. A water pipe which runs along the line $r = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\3\end{pmatrix}$ enters the green house roof at point P. The farmer wishes to move the sprinkler closer if the distance between S and P is more than 4m.

Two boundaries of the sprinkler spray zone on the green house floor have equations $y = 2x + 1$ and $3y + x = 12$. The farmer wishes to change the nozzle if the angle between the two spray boundaries is less than 45°

A faulty nozzle sprays water such that it hits the floor at a point $Q(x, y)$ whose distance from the point $A(0,1)$ is twice its distance from the point $B(3,7)$. The farmer wishes to relocate his pawpaw plant at (5,6) if water can't reach it.

Task:

Based on mathematical justifications, advise the farmer on the decisions to take.

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ITEM FOUR

An engineering team is installing a surveillance system on a school compound. Two straight support cables are fixed in space to hold a camera. The cables are modeled by the lines:

$L_1: \boldsymbol{r} = \begin{pmatrix}1\\2\\0\end{pmatrix} + \alpha\begin{pmatrix}2\\-1\\2\end{pmatrix}$ and $L_2: \boldsymbol{r} = \begin{pmatrix}3\\0\\4\end{pmatrix} + \beta\begin{pmatrix}1\\1\\-2\end{pmatrix}$.

The camera will be placed at a point where the two cables intersect.

A third power line runs along $L_3: \boldsymbol{r} = \begin{pmatrix}0\\1\\1\end{pmatrix} + \mu\begin{pmatrix}1\\2\\2\end{pmatrix}$. For safety, the engineering team wishes to install the system if the camera is at least 3m from the power line otherwise the support cables may be relocated.

The light beam from the camera traces a path on the ground. The cross-section of the beam is a parabola with equation $y = 2x^2 - 8x + 5$ where x and y are distances in metres from the base of the camera pole and the camera pole lies on the point of intersection of the axis of the parabola and x-axis. The team wishes to adjust the camera pole for wider view if the distance of the camera pole from the vertex is less than 3m.

Glare (very bright light) occurs when the product of sunlight intensity factor $2sin\theta - 1$ and reflection factor from the lens $sin\theta - 1$ is zero for $0° \leq \theta \leq 180°$. The engineering team wishes to program the camera to switch off if one of the angles is $150°$ to avoid glare.

Task:

Based on mathematical justifications, advise the engineering team on the decisions to take.

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ITEM FIVE

A certain city has completed construction of a new city hall complex. Before official opening, the city engineer is asked to carry out a technical audit of three features and make recommendations to the city council.

First, a sloped solar panel roof is to be installed over the main hall. The contractor has fixed three support pillars to hold the roof at points $A(4,2,18), B(12,6,21)$ and $C(6,14,19)$ in metres from the main gate, where $x =$ metres east, $y =$ metres north,

$z =$ metres above the ground level. A telecom mast stands nearby with its top at $T(8,8,24)$.

City safety regulations state that the roof must be at least 2.5 m below any nearby tall structure to reduce lightning-strike risk and allow crane access.

The engineer must determine if this roof design meets the safety clearance rule.

Second, in the courtyard, the landscaper designed a flower fountain. The edge of the fountain is set using:

$x = 8cost,\ y = 5sint\ \ 0 \leq t \leq 2\pi$ where x and y are in metres from the fountain center.

A visitor's bench has been placed 7m east and 2m north of the center. The council's policy is to relocate any bench that lies where water splash can reach it. The engineer must determine if the bench is safe from splash.

Third, a telecom company will mount a rooftop antenna. Their engineers report that signal noise is strongest at angles θ; where $5cos3\theta + 12sin3\theta = 13,\ 0° \leq \theta \leq 180°$. The antenna is currently planned to point at a bearing of 30°. The engineer must determine if the antenna bearing must be changed to avoid interference

Task:

Using mathematical evidence, guide the engineer on the decisions to take.

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ITEM SIX

A certain community is inspecting its main stadium before regional games. The technical assistant has been asked to guide and give clear decisions to the local organizing committee.

A new flat sunshade is to be built over the main stand. To drain water properly, the sunshade must be perpendicular to a support rod with direction $\begin{pmatrix}1\\2\\2\end{pmatrix}$. One corner of the shade must be fixed at post $P(2,4,10)$ in metres from the main gate, where x = East, y = North and z =height above the ground. A camera drone will fly from point $D(8,10,16)$ to film point P.

Broadcast rules state that the camera line of sight DP must not be with in 15° of the sunshade surface to avoid camera glare (too bright light) and obstruction. The technical assistant must state if the drone's path meets the broadcast safety rule.

The contractor also installed three new marker poles on the field at points $A(1,7), B(7,9)$ and $C(9,3)$, in metres from the center mark. He claims these poles lie exactly on the painted circular boundary line for the running track. Accepted sports rules require that such a circular track must have a radius of at least 36.5m to give runners enough space on bends. The technical assistant must state if the track is wide enough for accepted sports rules.

To broadcast the games, a straight cable is to run from hilltop M to hilltop N across a valley. From the control point at R, measurements show that RM = 200m, RN = 160m and angle MRN = 60°. The cable company will reject the job if the cable length MN is more than 190m, or if the angle of the cable from M drops below 40° to the line MN. The technical assistant must state if the cable plan should be approved or rejected.

Task:

Based on your mathematics skills, advise the technical assistant on the decisions to take.

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ALGEBRA

ITEM SEVEN

The academic committee of a certain school is planning inter-house quiz and cultural day. The deputy head teacher who is the chairperson of the committee wishes to justify the budget, seating plan and team selection. The committee has **shs 470,000** for **3** quiz books, **2** traditional attire sets and **4** trophies.

1 set of attires and **1** trophy together cost **shs 60,000** more than **2** quiz books. A trophy costs **shs40,000** more than a quiz book.

The committee has been given extra **shs150,000** and wishes to get more **two** sets of attires and **1** quiz if the extra money given to them is enough.

The main hall to be used has **12** rows for students. The first Row has **20** seats. Each row after that has 3 more seats than the row Infront. The committee will use the main hall if it can accommodate at least **400** students otherwise the committee can hire tents.

The committee also wishes to hire rectangular tables. According to the supplier, the cost $C(x)$ of one table is given by $ax^3 + bx^2 + 5000$ where x is the table width in metres. For **1 m** wide table, the cost is **shs7000** and for a **2 m** wide table, the cost is **shs 19,000**. The committee has allocated **shs50,000** for **3 m** wide table and wishes to hire it if the money allocated for it is enough.

Task:

Advise the committee with mathematical justification on the decisions to take.

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ITEM EIGHT

A coffee exporting company is preparing its strategic plan for European union buyers. The plan started at the beginning of **2024** and will end at the end of **2033**.

The company began replanting in **2024** with **12,000** new arabica seedlings. The board policy requires that planting increases by **1500** seedlings every year to meet the rising European union demand.

The company wishes to widen its market to Asia if by the end of **2033** it may have planted at least **180,000** seedlings.

The company also wishes to improve the livelihoods of their neighbors by giving out some seedlings. The number of seedlings given out in hundreds is given by

$\mathbf{10}(\boldsymbol{log_5 n + log_n 5})$. Where $\boldsymbol{n}$ is the number of years after **2024**. The company wishes to maintain the same number of seedlings it gives out if more than **2,500** seedlings are given out by **2026**.

The company's annual profit $\boldsymbol{P(t)}$ from coffee sales in millions of shillings is $\boldsymbol{at^3 + bt^2 + 5t + 20}$. Where $\boldsymbol{t}$ is the number of years after **2024**. The company made a profit of **29** million shillings in **2025** and in **2026**, the profit is expected to be **48** million shillings. The company wishes to increase on the number of its workers if the profit in **2033** exceeds **700** million.

Task:

Use mathematical reasoning to advise the coffee exporting company on the decisions to take.

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ITEM NINE

A certain financial organization is giving out an agriculture grant to local community groups. According to the grant officer, for a group to qualify for the grant, they must have a minimum number of bags of fertilizers and irrigation kits. For x bags of fertilizers and y irrigation kits, the sum of bags of fertilizers and irrigation kits must be at least 4.

A youth group had meetings in preparations for the application for a grant. In the first meeting they spent 5^{x+2} on fertilizers and 7^{y+1} on irrigation kits and they spent 3468 altogether. In another meeting, the amount 7^{y} was exactly 76 less than the amount 5^{x}. The group wishes to apply for the grant if they qualify.

The youth group has 8 men and 5 women. They want to form a committee of 5 people with at least two women to manage the grant. They will increase the committee size if the number of possible committees is fewer than 800.

The youth group has been dealing in maize and would wish to use the grant to boost maize production. The maize yield follows an arithmetic progression. The yield in third week was 11 bags and the yield in the fifth week was 17 bags. The group will to approve the funding if the total yield for the first 10 weeks exceeds 180 bags.

Task:

Based on mathematical skills, guide the youth group on the decisions to take.

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ITEM TEN

A construction company is designing a stepped viewing platform for a stadium. The heights of the steps above the ground form an arithmetic progression. The height of the eighth step is twice the height of the third step. The total height of the first eight steps is 39. The engineers wish to redesign the platform if the total height of the first three steps is greater than 9m.

The acoustics(sound) team wishes to reduce echo in the stadium by placing two sound meters a distance of x in tens of metres from the stage. One sound meter shows a reading of $(log_4 x)^2$ and another meter shows a reading of $2log_2 x$.

For people to hear clearly without echo, the sum of the two readings must be 3. The acoustics team wishes to adjust the sound meters if their distance from the stage is less than 20m

The electrical team wishes to position at least four lights. The main light is positioned at $-8 + 8\sqrt{3}i$ on the stadium floor plan. The other three lights must be positioned at the cube root of the main light position. The electrical team wishes to relocate the positions of the lights if the sum of the moduli of the positions of these lights is less than 20m

Task:

Based on mathematical evidence, advise the company on the decisions to make.

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ITEM ELEVEN

A certain youth group is fencing a rectangular vegetable project in a garden whose total area is 552 m^2. The garden is bounded by an existing wall along one longer side and another existing wall along one shorter side, so only the other two sides need to be fenced. The two fenced sides meet at a corner and they share one pole at that corner.

For the longer side that needs fencing, the group uses x poles and 2 metres of that side is used for tying the gate at one end. For the shorter side that needs fencing, they also use x poles but a space of 3 metres has a rock and no pole can be fixed there. The poles are separated by 2 metres. Part of the garden is to be used for banana plantation and it is not to be fenced. The group wishes to reduce the size of banana plantation garden if the number of poles used are greater than 30.

To irrigate the garden, they wish to build a water tank stand using y (in tens) concrete blocks. The height from the ground to the top of the stand is $\sqrt{y+25}$ metres and the height from the ground to the water pipe outlet is $\sqrt{y}$ metres. The pump requires the pipe of the outlet to be exactly 3m below the top of the stand. They wish to buy more concrete blocks and increase the height of the top for higher pressure if the blocks used are less than 100.

The group wishes to borrow money from the SACCO payable in 12 months. The SACCO uses binomial expansion to set repayments. The repayment for the n^{th} month is given by the n^{th} term in the binomial expansion $300{,}000(1+3x)^{11}$ where x is the multiplier and 300,000 is their initial deposit. The SACCO can give a top up loan if the they have paid back at least shs 500,000 with a multiplier of 0.02 in the first three months. They wish to apply for the top up if they qualify for it.

Task:

Using your mathematics skills, advise the youth group on the decisions to take.

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ITEM TWELVE

An investment group runs some projects to raise money for their wellbeing. It operates a poultry farm where its average monthly profit in millions of Ugx. after t months of operation is $\frac{2t^2+7t+1}{t^2+3t+2}$.

The group will expand this project only if there is any whole month when the average monthly profit exceeds 2 million, otherwise they will close it.

The group also operates a solar phone charging booth where the number of phones charged per day after t days is $4^{t+1} - 6(2^t) + 5$. The group will carry out a safety inspection on the day the booth charges exactly 45 phones. If that day is within the first five days of operation, they must replace the battery immediately because it is degrading too fast.

The group formed a SACCO to encourage savings. If a member deposits any amount at every start of the year, the SACCO gives annual compound interest of 5%.

A member qualifies for a token of appreciation if the total amount saved after 5 years exceeds UGX 5 million.

A certain member deposits Ugx. 200,000 at the start of every year. The group wishes to give this member a token of appreciation if he qualifies for it.

Task:

Basing on mathematical justifications, advise the group on the decisions to take.

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CALCULUS

ITEM THIRTEEN

A certain father has bought **18m** of wire mesh which is **1 m** wide. He has asked his son to use the wire mesh and make a cuboid chicken coop with a square base for his s.6 vacation poultry project.

The wire mesh must cover the four outer vertical sides and the top only. The bottom should be made of concrete. A wooden door of area **0.32 m^2** should be fitted on the front face. The son wishes to maximize the internal volume of the coop so it houses as many chickens as possible. The father wishes to widen the coop and buy more wire mesh if the capacity of the coop is less **$7m^3$.**

After approaching the supplier of the chicken, the rate at which chickens in hundreds are supplied per week is given by $t\sqrt{2t+3}$ where t is the time in weeks. The father will increase on the number of chickens bought if less than **400** chickens will be bought after the first two weeks of the vacation.

The rate of consumption of feeds in kg per week is also given by $50te^{0.2t}$ where t is the time in weeks, the father will increase on the capacity of feeds if more than **140 kg** are consumed after the first two weeks of the vacation.

Task:

Based on mathematical evidence, advise the father on the decisions to take.

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ITEM FOURTEEN

A school farm is rearing fish in a circular pond of depth 1.5m. The pond is being drained for cleaning and then refilled. The rate of change of the volume of water V m^3 in the pond, t hours after draining starts is $\frac{(3V+12)}{(t+1)(t^2+1)}, t \geq 0$.

Initiälly the pond contains 20 m^3 of water. The manager wishes to increase on the number of fish in the pond if after 3 hours the volume is more than 200m^3 of water.

The manager also noticed that an error of 0.02m was made in the radius of the pond which is supposed to be 2.5m. He wishes to engage the technicians for restructuring of the pond if this error in radius causes an error in volume greater than 2%.

Task:

Based on mathematical evidence, advise the farm manager on the decisions to take.

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ITEM FIFTEEN

A certain municipal council is planning to build a new storm water drainage system. In consultations with the municipal engineering team, the shape of a main pipe bend follows a curve and structural tests show that at every point on the pipe, the quantity $x^2 - 2x + 2y^2$ must be equal to the quantity $3xy + 4$. Where x represents distance in metres east of a manhole and y is the vertical depth in metres below the ground level. Inspection is done at the point $(1, -1)$.

For structural safety, a perpendicular support line must be installed at that point. The city council will only approve it if this support line goes through the ground base at $(4, 0)$, otherwise the design is unstable and the municipal council must revise it.

The rate of flow y m^3/s varies with time t seconds according to the law: The rate of change of flow plus twice the product of time and flow equals twice the time. Initially when the pumps are switched on, the flow is 0 m^3/s. The drainage gates are designed to open automatically only if the flow rate reaches at least $0.8m^3/s$ within the first 2 seconds, otherwise the municipal council must activate a manual override system to prevent flooding.

The volume of water in thousands of litres that passes through a filter in the first $\frac{\pi}{2}$ seconds is given by the total amount accumulated from the flow rate of $sint\sqrt{cost}$ litres per second. The treatment plant can only handle 600 litres in that period. If the calculated volume exceeds 600 litres, the municipal council must build an extra holding tank.

Task:

Using mathematics to justify your answers, advise the municipal council on the decisions to take.

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ITEM SIXTEEN

A certain village council is constructing an underground right circular cylindrical tank with an open top to store water for the village. The total internal surface area to be lined with water proof cement is fixed at 150π m^2. The village council wishes to get water to the maximum capacity but can relocate the tank if the depth exceeds 8m due to sandy soil.

The council wishes to use an electric pump which delivers water at a constant rate of 2π m^3. The council wishes to use another pump if the rate at which water is rising is less than 0.12 m per minute.

The cost of electricity to run the pump increases as the tank fills. The instantaneous cost of pumping is given by $h^2 ln2h$ in tens of shillings per metre^3 of water where h m is the current depth of water.

The council wishes to collect more money from the village members if the cost of filling the tank from a height of 1m to the maximum height is greater than shs500,000.

Task:

Based on mathematical evidence, guide the village council on the decisions to take.

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ITEM SEVENTEEN

A newly constructed stadium has an athletic track consisting of a rectangle with semicircles at each end. The rectangular section measures L metres long and $2r$ metres wide where r is the radius of each semicircle in metres. The perimeter of the track is 800m.

The head coach of an athletics team wants to use the rectangular field for team training drills. For effective training, the field must have at least $20{,}000\text{m}^2$. If the available area is less than this, the team will have to train elsewhere.

Besides the track, the coach is managing costs. The total weekly cost $C(n)$ in Ugx of running the team is $3n^2 + 200$, while the total performance score $P(n)$ of the team is $50n$ where n is the number of athletes selected. The coach defines the cost efficiency $E(n)$ as the ratio of $P(n)$ to $C(n)$ performance points per shilling. The team board will only approve the program if the cost efficiency can reach at least 0.08

In addition, the coach is also selecting athletes for upcoming hill races. During training, he studied one athlete and found that when running uphill, the athlete's speed in km/h is $2x^2 + 4x + 11$ where x is the distance in km covered after t hours, with $x = 0$ when $t = 0$.

The coach has set a selection criterion: If the athlete can cover at least 1.5 km in the first $\frac{\pi}{30}$ hours of hill running, then he will be selected for the hill race team, otherwise he will not be included.

Task:

Based on mathematical reasons, advise the coach on the decisions to take.

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ITEM EIGHTEEN

Stephen bought a wall paper roll that is 2m wide and 20m long. He wants to decorate his sitting room ceiling and one wall where the T.V will be placed. The ceiling is a square. The wall is rectangular with area $30m^2$, and one side of the wall is equal to the side of the ceiling. The ceiling has a central light fitting covering $0.8m^2$. He will leave this space empty when pasting wall paper. On the wall, a space of $2.4m^2$ must be left out for the T.V.

To reduce costs on decorative edge trim, Stephen wants the total perimeter of the ceiling and the wall to be as small as possible. He wishes to buy an extra roll if the wallpaper he has is not enough to cover both the ceiling and the wall.

Stephen pastes a wall paper at a rate $\frac{dW}{dt} = \frac{t+2}{\sqrt{t^2+4t+8}}$ where W is the area in square metres fixed after t hours with $W = 0$ when $t = 0$. He will hire another decorator if he pastes less than $3m^2$ in the first 3 hours.

For the central light on the ceiling, he wants to install a decorative panel whose shape is the region bounded by $y = 4 - x^2$ and $y = x^2 - 2x$. He will proceed with this panel only if its area does not exceed $1m^2$, otherwise it will be too large for the ceiling.

Task:

Using mathematics to justify your answers, advise Stephen on the decisions to take

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Item 1

A small courier business operating between Kabale and Mbarara uses motorcycles and small vans to transport parcels. The owner introduced a computerized system to estimate parcel charges, monitor fuel usage, and determine container capacities. However, he is not sure whether the calculations are reliable enough for business decisions.

Previous records showed that parcels of masses 200 g, 400 g, and 600 g were charged shs 700, shs 1200, and shs 3000 respectively. One customer brought a parcel weighing 185 g, while another claimed to have paid shs 2200, although the parcel mass was not recorded. The owner will keep the computerized charging system if the charge for the 185 g parcel is less than shs 700, and the estimated mass corresponding to shs 2200 lies between 400 g and 600 g; otherwise, the fare estimates will be reviewed before use.

The business also manufactures conical metallic fuel containers for motorcycles. One container had radius 4.7 cm and height 12.65 cm, measured with possible errors of ± 0.3 cm and ± 0.45 cm respectively. The owner will use the container measurements for production if the difference between the upper and lower limits of the possible volume does not exceed 140 cm^3; otherwise, the container dimensions will be remeasured.

In addition, the fuel consumption of one motorcycle while climbing a hilly road near Kabale was represented by the function $\tan x$ over the interval

$$\frac{\pi}{6} \leq x \leq \frac{\pi}{3}.$$

The transport officer used 6 ordinates to estimate the total fuel used during the climb. The transport officer will use the fuel estimate for planning if the percentage error is less than 5%; otherwise, a more accurate numerical method will be required.

TASK

As a student of Mathematics, help the business owner interpret the information, choose suitable methods, and make justified decisions about the charging system, container measurements, and fuel estimate.

Item 2

A computing technician is designing a calculator routine for finding the reciprocal of the fourth root of a positive number N. To improve efficiency, the routine is expected to begin with an approximate value obtained from nearby tabulated values and then refine the result using an iterative process.

During calibration, the following values were available:

N	15.0	15.2
Reciprocal of the fourth root of N	0.5081	0.5064

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The technician wants to determine the reciprocal of the fourth root of 15.1. If this quantity is denoted by x, the routine should start with a suitable approximation from the table correct to one decimal place.

The technician should then form an equation in x, derive an iterative formula, and obtain a value correct to 3 decimal places. The iteration will be considered stable if two successive approximations differ by less than 0.0005. The technician will install the routine on the calculator if this condition is satisfied; otherwise, the iteration formula will be revised.

The calculator is also being tested on the expression

$$Q = \frac{abc}{a - b},$$

using the recorded values

$$a = 5.15, \qquad b = 14.5, \qquad c = 8.50.$$

Since these are measured values, the technician wants the possible limits of Q to be examined carefully. The computed value of Q will first be recorded from the given measured values. The technician will approve the calculator output if the greatest possible error from the computed value of Q is less than 1; otherwise, the measurement procedure will be checked again.

TASK

As a student of Mathematics, help the technician interpret the information, choose suitable methods, and make justified decisions about the calculator routine and output reliability.

Item 3

An engineer working with a mobile communication company in Uganda is designing a signal synchronization system to improve network stability along the Kabale–Mbarara transmission line. The operating condition of the system is modelled by

$$\sin x = \ln x.$$

From preliminary observations, the engineer established that the required operating point lies within the interval

$$2 \le x \le 3.$$

To reduce signal delays and network interruptions, the engineer first wants an approximate value of the operating point using graphical method correct to 1 decimal place, followed by a more accurate value correct to 3 significant figures using a suitable iterative method. The engineer will use the operating point for synchronization if successive approximations converge to the same value; otherwise, the iterative model will be changed.

The engineer also studies the behaviour of the system over a suitable interval from the graph by considering the area under the curve

$$y = \sin x.$$

To estimate this area, tables of values are prepared at equal intervals using 5 equal strips. However, the engineer is not sure whether the estimated area is sufficiently accurate for system calibration.

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The engineer will approve the system for operation if the difference between the exact area under the curve

$$y = \sin x$$

and the estimated area is less than 0.05; otherwise, the calibration will be repeated.

TASK

As a student of Mathematics, help the engineer interpret the information, choose suitable numerical methods, and make justified decisions about synchronization and calibration.

Item 4

A transport records officer is preparing a fare and accounting report for a taxi company operating between Kampala and Mukono. The taxi stages along the route are 5 km apart, and the taxi stops only at these stages except in emergencies. The fares from Kampala to the 1st, 2nd, 3rd, and 4th stages are Shs 1200, Shs 1700, Shs 2100, and Shs 2600 respectively.

During one journey, a passenger travelling to Mukono fell sick and was dropped at a clinic located 3 km before Mukono. Another passenger with only Shs 1900 was allowed to travel and had to be dropped at a point equivalent to the amount paid. The officer will keep the fare system if the estimated fare reduction for the sick passenger is at least Shs 200, and the passenger with Shs 1900 can travel at least 10 km from Kampala; otherwise, the fare table will be revised.

The officer is also comparing fuel records from two monitoring systems installed in the taxi. The recorded quantities are

$$p = 36.8, \qquad q = 8.47, \qquad r = 52.61, \qquad s = 4.28,$$

where each value has been rounded to the stated number of decimal places. To check whether the systems give consistent estimates, the officer uses

$$\frac{p}{q} - \frac{r}{s}.$$

The officer will rely on the monitoring system if the estimated percentage error in the computed result is less than 2%; otherwise, the fuel records will be rechecked.

At the end of the day, the taxi driver decides to save Shs 500 000 with a savings society in Kampala offering compound interest at 12% per annum compounded yearly. The driver wants the amount accumulated after 4 years together with a dry run representing the calculation process. The driver will keep the savings plan if the accumulated amount exceeds Shs 750 000; otherwise, a better investment option will be sought.

TASK

As a student of Mathematics, help the officer interpret the information, choose suitable methods, and make justified decisions about the fare system, fuel monitoring, and savings plan.

Item 5

A water engineer in Wakiso District is designing cylindrical water pipes for a small community irrigation project. One pipe has radius 2.5 cm measured to the nearest 0.1 cm and height

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80 cm measured to the nearest centimetre. The engineer wishes to determine the range wi which the exact volume of the pipe lies.

The engineer will use the pipe design if the difference between the upper and lower limits of possible volume does not exceed 3500 cm^3; otherwise, the pipe dimensions will be redesig to avoid insufficient or excess water supply.

During the same project, the engineer develops a mathematical model for regulating wa pressure in three sections of the pipeline. The pressure settings are represented by

$$x^3 - 6x^2 + 11x - 6 = 0.$$

The engineer is required to investigate the pressure model graphically on the interval

$$0 \leq x \leq 4.$$

A table of values should be prepared and the graph of

$$y = x^3 - 6x^2 + 11x - 6$$

drawn using a suitable scale. The operating pressure settings must be read from the poi where the curve cuts the x-axis, not by factorising the equation.

For safe operation, each pressure setting read from the graph must lie within

$$0 \leq x \leq 4.$$

In addition, the pressure setting read from the graph near $x = 3$ must differ from 3 by less than 0.01. If these conditions are not satisfied, the pressure regulator will be reset.

TASK

As a student of Mathematics, help the engineer interpret the information, choose suitable methods, use a graph to estimate the pressure settings, and make justified decisions about the pipe design and pressure system.

SECTION B (PART I)ITEMS

Item 6

The management committee of a secondary school became concerned about learners' academic performance, staff punctuality, and the safety of the school transport system after several complaints from parents and teachers. During a planning meeting, the committee resolved that future decisions should be guided by proper mathematical analysis rather than assumptions. They therefore wished to study examination performance in order to improve academic standards, evaluate the reliability of staff punctuality arrangements, and assess whether the road commonly used by the school van was safe enough for regular transport activities.

The marks obtained by learners in a mock examination were grouped as shown below.

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Marks Interval	Number of Students
$0 < x \leq 15$	6
$15 < x \leq 30$	8
$30 < x \leq 35$	15
$35 < x \leq 40$	20
$40 < x \leq 55$	10
$55 < x \leq 65$	7
$65 < x \leq 75$	4

The committee wished the Director of Studies to estimate the median mark using a suitable statistical graph and also use a computational method to determine the mean mark and standard deviation. The committee will maintain the current teaching plan if the mean mark is at least 50; otherwise, weekly tests will be introduced.

The committee also studied the punctuality of a staff member who travels by either route A or route B. The chance that he uses route A is 0.25. If route A is used, the chance that he arrives late is 0.6, while for route B it is 0.3. The committee will keep the current staff travel arrangement if the chance that the staff member used route B, given that he arrived late, is less than 0.7; otherwise, a stricter punctuality plan will be introduced.

In addition, accident records on a road used by the school van showed that the monthly number of accidents is rectangularly distributed from 5 to 12. The committee wished to determine the expected number of accidents in a month and the chance of having more than 7 accidents. The school will continue using the route for regular transport if this chance is less than 0.4; otherwise, an alternative route will be chosen.

Task

As a student of Mathematics, help the school management committee interpret the information, choose suitable methods, and make justified decisions about academic support, staff punctuality, and school transport.

Item 7

A fruit supplier sorting oranges for nearby schools placed two baskets on the same table. Basket A contained 7 yellow oranges and 3 green oranges, while Basket B contained 4 yellow oranges and 5 green oranges.

To improve the appearance of Basket B, one orange was randomly transferred from Basket A to Basket B. Later, two oranges were picked from Basket B one at a time without replacement and returned to Basket A. The supplier wished to know whether the colour balance in Basket A would remain satisfactory. The supplier will keep the sorting method if the chance that the transferred orange and the two returned oranges are all of the same colour is at least 0.20; otherwise, the sorting procedure will be changed.

The supplier also wished to study the number of green oranges in Basket A after the transfer process. The supplier will use the process for regular sorting if the expected number of green oranges lies between 3 and 5, and the standard deviation is less than 1; otherwise, the baskets will be sorted manually.

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In addition, apples are packed in sacks containing 20 apples each. Records show that the chance of selecting a rotten apple from a sack is 0.15. The supplier will continue packing apples in sacks of 20 if the most likely number of rotten apples in a sack does not exceed 4; otherwise the apples will be inspected more carefully before packing.

Task

As a student of Mathematics, help the supplier interpret the fruit-sorting and packing information, choose suitable methods, and make justified decisions.

Item 8

A quality control team in a mobile phone assembly company is checking whether its production and inspection procedures are reliable enough for customer supply. Some customers have complained about differences in component sizes and defective phones reaching the market.

In one department, the masses of bolts used in assembling phone components were recorded as shown below.

Mass (grams)	98	99	100	101	102	103	104
Number of bolts	8	11	14	20	17	6	4

The company will use the bolts for production if the mean mass differs from 101 g by not more than 1 g; otherwise, the bolt batch will be rejected.

In another department, the company imports two phone brands. In one shipment, 55% were Brand A phones and 45% were Brand B phones. The probability that a Brand A phone is defective is 0.04, while the probability that a Brand B phone is defective is 0.06. The company will continue importing both brands together if the probability that a randomly selected phone is defective does not exceed 0.05; otherwise, the import proportions will be reviewed.

To improve inspection, the quality control team models the number of faults detected in a phone by the probability function

$$f(x) = \begin{cases} k2^{-x}, & x = 1, 2, 3, \\ 0, & \text{elsewhere.} \end{cases}$$

The company will keep the current inspection process if the expected number of faults detected is less than 2.5; otherwise, the inspection method will be improved.

The management therefore wants a careful mathematical review before deciding on production, importation and inspection procedures.

TASK

As a student of Mathematics, help the management interpret the information, choose suitable methods, and make justified decisions about the quality control procedures.

Item 9

A school canteen is planning whether to continue selling packed fruit juice to learners during evening prep time. The canteen buys 15 packets of juice every day. From previous sales records,

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the probability distribution for the number of packets sold in a day is shown below.

Number of packets sold	10	11	12	13	14
Probability	0.10	0.15	0.25	0.30	0.20

Each packet is bought at shs.1800 and sold at shs.2500. Any unsold packets expire and are discarded. The canteen will continue stocking the juice if the expected daily profit is at least shs.5000; otherwise, the juice business will be stopped or reduced.

To promote fairness during customer rewards, two sealed boxes are prepared. Box A contains 5 blue and 3 yellow coupons, while Box B contains 4 blue and 4 yellow coupons. One box is selected at random and two coupons are picked from it without replacement. The canteen will use the reward system if the probability of picking two yellow coupons does not exceed $\frac{1}{4}$; otherwise, the coupon arrangement will be changed.

The canteen management also observes that a learner who buys juice has probability 0.4 of returning the next day for another purchase. During one week, 8 learners are selected at random. The canteen will continue the sales campaign if the probability that at least 3 learners return exceeds 0.55; otherwise, a new campaign strategy will be designed.

TASK

As a student of Mathematics, help the manager interpret the situation, choose suitable methods, and make justified decisions about the juice business, reward system, and sales campaign.

Item 10

A football academy is reviewing the performance of its senior team before enrolling it into a regional tournament. The team is scheduled to play three important matches against different opponents. From previous performance records, the probabilities of winning the first, second and third matches are estimated to be $\frac{3}{5}$, $\frac{2}{5}$ and $\frac{1}{5}$ respectively.

The academy management wants to know whether the team is likely to perform well enough to justify the cost of travelling and training. The academy will enter the team into the regional tournament if the probability of winning at least two matches exceeds 0.30; otherwise, tournament entry will be delayed.

To guide future planning, the management also defines a random variable X as the number of matches won by the team. The academy will keep the current training programme if the expected number of matches won is at least 1; otherwise, the training programme will be revised.

In another competition involving teams of nearly equal strength, the academy assumes that the probability of winning a match is 0.6. The team is expected to play 8 independent matches during the season. Management wants to estimate the likelihood that the team wins at least 5 matches. The academy will regard the season target as achievable if this probability exceeds 0.50; otherwise, the season plan will be reviewed.

The academy is therefore uncertain whether the team is prepared well enough for the competitions and wants a careful mathematical analysis before making final decisions.

TASK

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As a student of Mathematics, help the academy interpret the information, choose suitable methods, and make justified decisions about tournament entry, training, and season planning.

Item 11

A beverage company in Mbarara is testing the reliability of some production and recruitment decisions before introducing a new factory line.

The lifetime X, in years, of a machine component is modelled by the probability density function

$$f(x) = \begin{cases} \beta, & 2 < x < 3, \\ \beta(x-2), & 3 < x < 4, \\ 0, & \text{otherwise.} \end{cases}$$

Management will buy the component in bulk if the median lifetime exceeds 3 years; otherwise, a more durable component will be sought.

The marketing department is studying two independent customer choices: buying mango juice and buying passion fruit juice. The probability that a customer buys both is $\frac{2}{15}$, while the probability that a customer buys at least one of them is $\frac{3}{5}$. A flavour will be retained on the production line if its individual buying probability is at least 0.30; otherwise, it will be redesigned or removed.

During recruitment, each trainee performs 7 independent machine-start trials. From past records, a correct start is twice as likely as an incorrect start. A trainee will proceed for further training if the probability of making exactly two incorrect starts is greater than 0.25; otherwise, the trainee will be given remedial practice first.

Task:

As a student of mathematics, help management interpret the models, choose suitable methods, and make justified decisions on component purchase, flavour retention, and trainee selection.

Item 12

A school health committee and a small packaging company are reviewing data before making planning decisions.

In the school, the heights and masses of ten students were recorded as shown below.

Height (cm)	156	151	152	146	160	157	149	142	158	141
Mass (kg)	62	58	63	58	70	60	55	57	68	56

The committee wants to study the relationship between height and mass using a scatter diagram, a line of best fit, and rank correlation. The school will use height to estimate a student's mass for uniform planning if the rank correlation coefficient shows a positive relationship of at least 0.60; otherwise, actual masses will be measured before ordering uniforms. The committee also wants to estimate the mass corresponding to a height of 155 cm.

At the packaging company, the masses of soap powder packets are normally distributed with mean 842 g and variance 225 g^2. A random sample of 120 packets is selected. The company

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will keep the current filling machine setting if the probability that the sample mean mass lies between 844 g and 846 g is less than 0.10; otherwise, the machine will be checked for possible overfilling.

In another production line, experience shows that 8% of packets are underweight. During inspection, 200 packets are selected at random. The company will accept the production line if the probability that at most 20 packets are underweight, estimated using a normal approximation to the binomial distribution, is at least 0.80; otherwise, the filling process will be adjusted.

Task

As a student of Mathematics, help the school health committee and the packaging company interpret the information, choose suitable mathematical methods, and make justified decisions about uniform planning, machine filling, quality control, and production-line acceptance.

SECTION B (PART II)ITEMS

Item 13

A school workshop is being prepared for a science exhibition. A heavy demonstration machine of mass 18 kg with a square base $ABCD$ has to be moved from the store to the display area without damaging the tiled floor. The side AB of the square base is taken as the positive x-axis, while the side BC is taken as the positive y-axis.

To test whether the machine can first be moved safely by pushing along the sides of its square base, four students apply forces of 18 N, 22 N, 10 N, and 14 N along AB, BC, CD, and DA, respectively. The single force equivalent to the four applied forces will be approved as efficient only if its magnitude exceeds 10 N.

The machine is then loaded onto a trolley of mass 8 kg. To reduce the risk of the trolley front wheels jamming, the loaded trolley is pulled by a handle using a force of 120 N inclined at 25° above the horizontal. On the rough workshop floor, its speed increases from rest to 4.8 m s^{-1} in 4 s. This method will be allowed only if the coefficient of friction is not greater than 0.50, since a higher value would indicate too much rubbing and possible damage to the tiled floor.

For better control, the teacher considers another method. The 18 kg machine is placed on a trolley of mass 12 kg, while another trolley of mass 8 kg carries a 10 kg box of display materials. The two loaded trolleys are connected by a light inextensible rope and move along a smooth corridor floor.

The trolley carrying the machine is placed in front and is pulled by a horizontal force of 75 N. As it moves forward, it pulls the rear trolley carrying the box of display materials through the connecting rope. This method will be approved only if the acceleration does not exceed 2.8 m s^{-2}, to avoid sudden movement of the machine.

Task

As a student of mathematics, interpret the situation, choose suitable methods, and use your results to judge the safer method of moving the tiled machine.

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Item 14

A school is preparing an upstairs classroom for a parents' meeting, but the staircase is crowded. Some files and small teaching materials are needed quickly, so the teacher considers different safe ways of moving them.

A bag of teaching materials of mass 8 kg is suspended from a fixed beam at point A by a light string of length 50 cm. It is pulled sideways and kept at rest 30 cm from the vertical line through A, so that it does not rub against the wall. The teacher wants to know the tension in the string and the horizontal pull P. This method will be accepted only if the tension does not exceed 100 N.

Two groups of learners are also carrying extra chairs and stationery towards the same block. The first group moves at 8 km h^{-1} in the direction $N80°W$, while the second moves at 5 km h^{-1} in the direction $S25°W$. The teacher wants to know their resultant speed and direction. Their movement will be considered safe only if the resultant speed does not exceed 10 km h^{-1}.

Since the staircase is still crowded, the teacher tests whether a soft bag of files can be thrown from the ground to a wide basket on the balcony. The balcony is 6 m high and 18 m horizontally from the throwing point. The bag is thrown at 20 m s^{-1} at 40° to the horizontal. The teacher wants to know the time taken to reach the balcony position and the height of the bag there. The throw will be accepted only if the height is between 5.5 m and 6.5 m.

Task

As a student of Mathematics, analyse the situation and advise the teacher on the safest way to move the materials.

Item 15

At 12 noon, a traffic officer is monitoring two vehicles near a busy junction. The police post is taken as the origin. Vehicle A is a school van and vehicle B is a delivery truck. Their positions and velocities are recorded as

At 12 noon, the position vectors of vehicles A and B are

$$\mathbf{r}_A = (5\mathbf{i} + 2\mathbf{j}) \text{ km} \quad \text{and} \quad \mathbf{r}_B = (7\mathbf{i} + 7\mathbf{j}) \text{ km},$$

while their velocity vectors are

$$\mathbf{v}_A = (15\mathbf{i} + 10\mathbf{j}) \text{ km h}^{-1} \quad \text{and} \quad \mathbf{v}_B = (9\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}.$$

The officer wants to know whether the two vehicles will reach the same point if they continue moving with their present velocities. If they meet within the next 30 minutes, the officer must immediately warn one driver to slow down or change route. The officer also needs the time and position of the possible meeting point so that help can be directed to the correct place.

Later, vehicle B breaks down near the junction and must be towed away from the road. A police vehicle of mass 1800 kg tows the damaged vehicle of mass 1200 kg using a light inextensible rope along a rough horizontal road. The police vehicle applies a driving force of 6000 N. The damaged vehicle experiences friction with coefficient 0.20, while the police vehicle experiences a resistance of 800 N.

The towing method will be accepted only if both the motion remains controlled and the rope is not overstrained; that is, the acceleration must be less than 2 m s^{-2} and the tension must

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not exceed 3500 N. If this condition is not satisfied, the officer must call a breakdown truck instead.

Task

As a student of Mathematics, analyse the situation and advise the traffic officer on the risk of the vehicles meeting and whether the police vehicle can safely tow the damaged vehicle away.

Item 16

A school workshop is preparing materials for a science exhibition. Some boxes have to be moved up a wooden ramp to the display room, and a temporary direction sign has to be fixed at the entrance to guide visitors.

A box A of mass 3 kg rests on a rough ramp inclined at 30° to the horizontal. It is connected by a light string passing over a smooth pulley to a hanging box B of mass 2 kg. When the boxes are released from rest, box B moves downwards with an acceleration of 0.5 m s^{-2}. The teacher wants to know the coefficient of friction between box A and the ramp. If the coefficient of friction is less than 0.20, the ramp must be covered with a rough mat before boxes are moved on it.

At the exhibition entrance, a small direction sign is supported at one point by three ropes. One rope pulls horizontally to the right with a force of 15 N. A second rope pulls upwards and to the right with a force of 20 N, making an angle of 50° with the horizontal. A third rope pulls downwards and to the left with force P, making an angle θ below the horizontal, so that the sign remains at rest. The teacher wants to know the magnitude of P and the angle θ. If P is greater than 30 N, an extra support must be added to the sign before visitors arrive.

Task

As a student of Mathematics, analyse the situation and advise the teacher on whether the ramp is safe for moving boxes and whether the sign can remain safely supported.

Item 17

After heavy rain, health centre workers need to lower a 50 kg emergency oxygen cylinder from a raised store down a rough ramp inclined at an angle θ to the horizontal, where $\sin\theta = \frac{3}{5}$. The coefficient of friction between the cylinder and the ramp is 0.25.

To control the motion, the cylinder is tied to a light inextensible string which passes over a smooth fixed pulley at the top of the ramp, then under a smooth movable pulley of mass 30 kg, before being fixed to a strong hook above.

The lowering arrangement will only be trusted if the tension in the string is not greater than 115 N. However, the workers also want to know what would happen if the worn string snaps when the cylinder is still 2 m from the foot of the 8 m ramp. In that case, the cylinder should reach the floor with a speed less than 8 m s^{-1}, so that it can still be handled safely.

After reaching the foot of the ramp, the cylinder moves along a rough horizontal concrete floor towards a restroom wall 9 m away.Immediately it reaches the floor, a horizontal force of 80 N is applied on towards the wall. The coefficient of friction between the cylinder and the floor is 0.40. The restroom wall will be considered safe only if the cylinder comes to rest before reaching it.

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Task

As a student of Mathematics, help the workers interpret the motion, choose suitable methods for the different stages, and make correct judgement

Item 18

At a building site, workers need to move an 8 kg portable concrete vibrator up a rough wooden ramp into a temporary store. The ramp is inclined at 40° to the horizontal. To make the work easier and more controlled, the machine is attached to a light inextensible rope which passes over a smooth fixed pulley at the top of the ramp, then under a smooth movable pulley carrying a 15 kg counterweight, and the other end of the rope is fixed to a firm support. The coefficient of friction between the machine and the ramp is 0.20.

When the system is released from rest, the counterweight moves vertically downwards while the machine moves up the ramp. The foreman will allow this method only if the machine moves up the ramp in a controlled way and the rope tension does not exceed 90 N.

Later, two site vehicles A and B, carrying materials through different access routes, are monitored from a fixed point O. The drivers are not sure whether their routes only pass close to each other or actually meet at the same point. At noon, their position vectors and velocity vectors are

$$\mathbf{r}_A = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k} \text{ km}, \qquad \mathbf{v}_A = 6\mathbf{i} + 4\mathbf{j} + 2\mathbf{k} \text{ km h}^{-1},$$

and

$$\mathbf{r}_B = 8\mathbf{i} + 7\mathbf{j} + 3\mathbf{k} \text{ km}, \qquad \mathbf{v}_B = -6\mathbf{i} - 4\mathbf{j} - 2\mathbf{k} \text{ km h}^{-1}.$$

The safety officer must issue a radio warning if calculations show that the two vehicles will reach the same point at the same time before 12:40 p.m.

Task:

As a mathematics student, help the foreman interpret the situations, choose suitable methods, and judge whether the ramp-and-pulley arrangement is safe and whether a radio warning is needed.

Item 19

A site foreman supervising the construction of a two-level building wants workers to use a pulley system to raise building materials to the upper floor and lower waste materials safely to the ground. A 240 kg bag of aggregates B rests on a rough inclined plane AD, while a 380 kg waste skip C hangs vertically from the same light inextensible rope passing over a smooth pulley at A. The plane AD is 20 m long and rises through a vertical height of 12 m. The coefficient of friction between the bag and the plane is 0.15.

The foreman will allow the lifting operation only if the bag is pulled up the plane in a controlled way and the rope tension does not exceed 2600 N. He is also worried that if the rope breaks just as the bag reaches A, the bag may slide back down the rough plane. The return motion will be considered safe only if the speed of the bag on reaching D is less than 15 m s^{-1}.

After reaching D,to guide it towards a loading point near the fuel storage area, workers apply a constant horizontal force of 400 N in the direction of motion along a rough surface with

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coefficient of friction 0.25. The fuel storage area is 32 m from D. The workers are worried if the fuel storage area is safe from the bag.

Task:
As a mathematics student, help the site foreman analyse the motion, choose suitable methods, and judge whether the lifting operation, the return motion down the plane, and the fuel storage area are safe.

Item 20

A site engineer is supervising the installation of a rectangular steel roof frame $ABCD$ of mass 20 kg. The frame has dimensions $AB = 12$ m and $AD = 5$ m, taken along the positive x- and y-axes respectively. During positioning, workers apply forces of 5 N, 6 N, 3 N, 2 N, and 13 N along AB, BC, CD, DA, and AC respectively.

The engineer will approve the positioning method only if the single force equivalent to the five applied forces does not exceed 17 N. He also needs its direction so that workers know the safe guiding direction.

While being lifted, the frame accidentally slips from rest from a height of 8 m. The fall will be considered safe for handling only if its speed on reaching the ground is less than 13 m s^{-1}.

After reaching the ground, the frame continues moving along a rough horizontal surface towards a storage wall 50 m away. To guide it away from other materials, workers apply a constant horizontal force of 20 N in the direction of motion. The coefficient of friction between the frame and the surface is $\frac{1}{5}$. The wall will be considered safe only if friction is sufficient to slow the frame down and bring it to rest before reaching the wall.

Task:
As a mathematics student, help the site engineer interpret the situation, choose suitable methods, and judge whether the positioning method, the falling motion, and the storage wall are safe.

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